High-Resolution Sustain Pedal Depth Estimation from Piano Audio across Room Acoustics

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Introduction

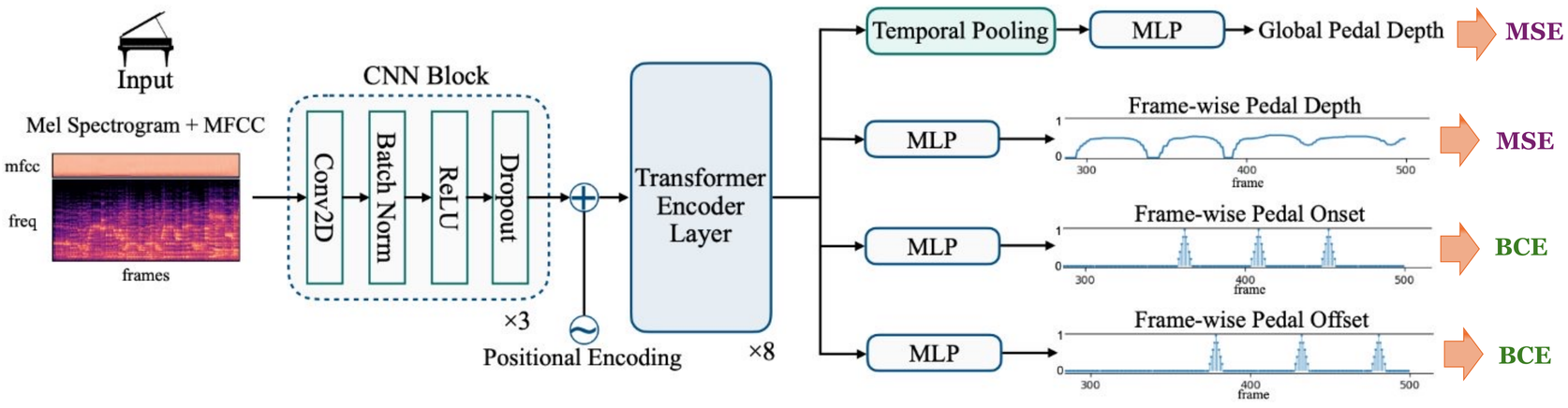
- Beyond on/off prediction → **continuous pedal depth**
- Evaluate under **varied room acoustics** (leave-one-room-out)
- **Synthetic audio dataset** with **multiple room configs**

Conclusion

- **Continuous pedal depth** beyond on/off
- **Finer, musically meaningful** analysis of pedaling
- **Robustness:** reverb can **overestimate** depth; **diverse rooms** help

Methodology

Model



Dataset

MAESTRO V3.0.0:

- Audio + Pedal Data

Synthetic Audio:

- MIDI from MAESTRO + Pianoteq 8 Stage

Room	Name	Mix	Dur	Size	PreD
1	Dry Room	-	-	-	-
2	Clean Studio	+10dB	0.4s	12m	0s
3	Concert Hall	+50dB	4s	50m	0.01s
4	Church	+10dB	2.5s	18m	0s

Table 1. Summary of room acoustic settings used for synthesized audio. Each room configuration specifies the reverb mix level (Mix), reverb duration (Dur), room size (Size), pre-delay (PreD), etc.

Experiment Results

Quantitative Analysis

Model	Binary			4-Class			MSE ↓	MAE ↓
	P ↑	R ↑	F1 ↑	P ↑	R ↑	F1 ↑		
Kong et al. [6]	0.9374	0.9374	0.9373	0.4752	0.6786	0.5588	0.0762	0.1579
Yan and Duan [8]	**0.9446**	**0.9445**	**0.9443**	0.4790	0.6836	0.5631	0.0710	0.1528
Ours (BCE)	0.8907	0.8911	0.8906	0.6039	0.6507	0.6180	0.0584	0.1537
Ours	0.8975	0.8971	0.8973	**0.6849**	**0.6971**	**0.6863**	**0.0425**	**0.1339**

Our model demonstrates:

(1) lower MSE and MAE, along with significantly higher 4-class scores
(2) binary detection performance comparable to baselines
(3) higher accuracy with error thresholds > 0.15

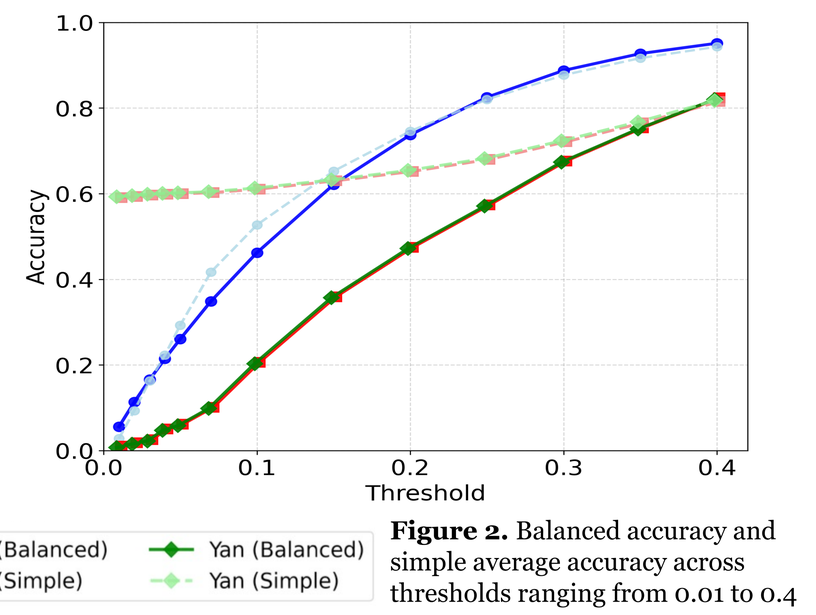
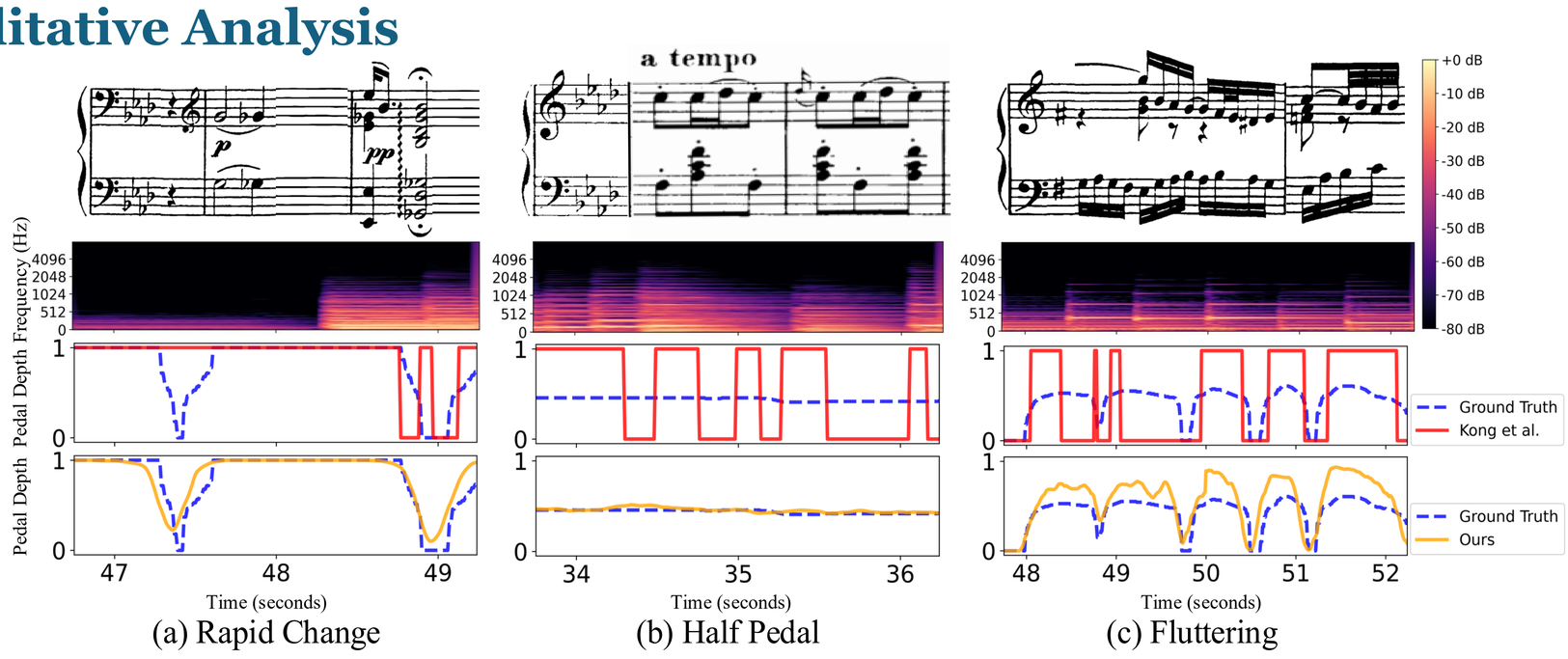


Figure 2. Balanced accuracy and simple average accuracy across thresholds ranging from 0.01 to 0.4

Qualitative Analysis



(a) Rapid Change (b) Half Pedal (c) Fluttering

Figure 3. Comparison of our predicted pedal curves and the baseline binary model [6] against ground truth pedal depth values, plotted alongside aligned mel-spectrograms and their respective scores:

(a) F. Chopin (1810–1849) Polonaise-Fantaisie Op. 61, mm. 6–7

(b) F. Schubert (1797–1828) Impromptu in F Minor, Op. 142 No. 4, mm. 45–46

(c) J. S. Bach (1685–1750) Prelude and Fugue in E Minor, BWV 855, mm. 11–12.

Pedal Prediction Robustness

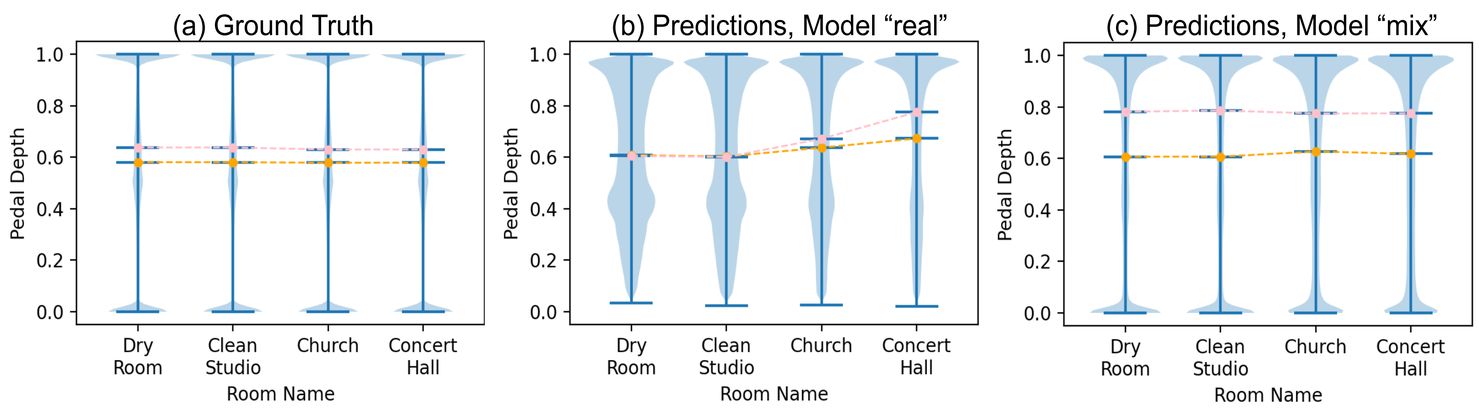
(1) In-domain vs OOD rooms: All models drop on rooms not seen during training

Model	Test Data				
	O	R1	R2	R3	R4
Kong et al. [6]	0.1579	0.1773	0.1780	0.2458	0.2247
Yan and Duan [8]	0.1528	0.2217	0.2195	0.2449	0.2365
Ours (BCE)	0.1537	**0.1654**	**0.1681**	0.2091	**0.1894**
Ours	**0.1339**	0.1889	0.1841	**0.2039**	0.1905

(2) Leave-one-room-out experiments: More room diversity → better generalization

Training Data	Test Data			
	R1	R2	R3	R4
R2+R3	0.1125	0.1119	0.1453	0.1654
R1+R3	0.1107	0.1145	0.1433	0.1713
R1+R2	0.0979	0.0981	0.1863	0.1779
R1+R2+R3	0.1062	0.1065	0.1439	0.1619

(3) Effect of room acoustics on pedal prediction



- Ground-truth depth stays *stable* across rooms
- Model trained *only on real recordings* shifts upward with stronger reverb
- *Mixed rooms* in training bring distributions back in line

Figure 4. Violin plots showing the distribution of predicted pedal depth values (b and c) and ground truth (a) across four rooms.